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Formulation and bio-efficacy of different isolates of *Beauveria bassiana* against adults and third nymphal instar of desert locust (*Schistocerca gregaria* Forskål)

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HIGHLIGHTS

- Climate change has induced episodic upsurge in desert locust (*Schistocerca gregaria*)
- There is increased use of entomopathogenic fungi for control of many insect species.
- *Beauveria bassiana* is used to control insects pests including the desert locust.
- Less information is available on bioefficacy of formulation desert locust control.
- The isolates and formulations *B. Bassiana* were effective against desert locusts.

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ABSTRACT

Recently, the Horn of Africa witnessed a swift increase in the incidence of desert locust (*Schistocerca gregaria*) invasion. During outbreaks, pesticides are applied through aerial or ground spraying to kill the insects, and/or to prevent their spread to new grounds. However, after decades of extensive use, many drawbacks such as contamination of the environment, toxicity to non-target organisms, harmful residues on food, pest resistance, and bioaccumulation in the food chains emerged. Entomopathogenic fungi offer viable alternatives to chemical pesticides against many insect invasions, but few studies have tested their bio-efficacy in desert locusts. Therefore, the current study aimed at isolating, formulating local isolates 231, 334, 333, 341, 349, 351, 339 of *Beauveria bassiana*, and testing their bio-efficacy against larval and adult desert locusts. The 21-day experiment was conducted under controlled conditions in a greenhouse. Soil samples were collected from two agroecological zones in Isiolo and Laikipia Counties in Kenya. *B. bassiana* was isolated from the soil samples using the *Galleria* bait method and cultivated in Sabouraud Dextrose Agar Yeast (SDAY). The isolates were identified based on molecular techniques (DNA and PCR amplification). The conidia of the isolates were screened and bioassays on 30 locusts was conducted for 14 days. The best isolates eliciting over 90 % mortality during screening were used for formulations using three carrier materials (liquid paraffin, Diatomaceous Earth, and whey) which were again tested against adult and 3rd nymphal instars of the locusts. The stability of the formulations was also tested after 1 and 2 months. All the tested isolates of *B. bassiana* significantly outperformed the control and thus pathogenic to the adults and 3rd nymphal instars of *S. gregaria* under laboratory conditions. They caused mortality ranging from 57.8–100 % after 14 days post-incubation. The isolates 341, 231, and, 334 elicited 50 % mortality responses at concentrations 1.1×10^5 conidia/ml, 2.5×10^5 conidia/ml and 1.7×10^6 conidia/ml respectively in adults and 1.1×10^5 conidia/ml, 2.5×10^5 conidia/ml and, 1.7×10^6 conidia/ml respectively in 3rd nymphal instars. Formulations with 341–1, 341–2, and, 334–1 had the highest efficacy (>99 %) against the adult locusts. There was a significant 3-way interaction ($P < 0.05$) of isolate for the formulation, carrier material and, time in determining the Cfu of the *B. bassiana* formulations. After 1 month, the best Cfu occurred in formulation with isolates 231 and 341 formulated using Diatomaceous Earth, while the highest Cfu was observed in formulation with isolate 334 formulated with either liquid paraffin or whey. After 2 months, the highest Cfu occurred in

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formulation with isolates 231, 341, and, 334 formulated using liquid paraffin. This study provides encouraging data for the development of potential biopesticides formulated with different carrier materials against desert locusts.

1. Introduction

The voracity of desert locusts (*Schistocerca gregaria* Forskål), is a major concern in agriculture and the environment due to the damage to crops and vegetation (Katel et al., 2021). On their migratory journey, the desert locusts consume vast amounts of vegetation, resulting in immense economic losses of food crops and vegetation in pasture lands (Chatterjee, 2022; Mongare et al., 2023). A flight swarm typically comprises up to 80 million locusts/km^2, capable of covering a distance of more than 100 km, and can sweep an area of about 1200 km^2 every day (Babar, 2023). In the wake of climate change and its long-term effects, desert locusts has occurred in huge magnitude in recent years than previous episodes and caused more damage than any time in the last 100 years (Hassan and Aslam, 2024; Wang et al., 2021). The most devastated areas are in Sub-Saharan Africa where there is already chronic food deprivation (Dong et al., 2023). Increased locust upsaurge has also been previously associated increased school dropouts in Africa (Asare et al., 2023; De Vreyer et al., 2012). Between 2019 and 2020, exceptional locust upsurge and breeding was observed in the horn of Africa, a magnitude larger than previous invasions in the last 70 years in East Africa (Kimathi et al., 2020; Yusuf et al., 2024). Therefore there is an urgent need to broaden the measures for desert locust control strategies to avoid such devastating instances.

Control of the outbreaks or spread of desert locusts largely relies on the use of chemical pesticides (Showler, 2019). The conventional chemical insecticides that have found intense use include organochlorines, organophosphates, carbamates, and pyrethroids (Mullié et al., 2023; Singh and Singh, 2022). Although an assortment of these chemical pesticides may provide an effective means of desert locust control, increased attention has been raised concerning their toxicity, environmental persistence and bioaccumulation in biota as well as in nontarget organisms (Athar et al., 2024; Lecoq and Cease, 2022). The continuous use of chemical pesticides in biological control against several pests has led increased resistance to the applied pesticides as well as entailing high costs for emergency control (Lu et al., 2022). More recently commentators have expressed concern at the sheer scale of interventions using pesticides as biological control against locust and grasshoppers during upsurge which involved the application of 1.4 million kg of pesticide dust (Mullié et al., 2023). Because of these effects, most of these pesticides were withdrawn from the market leading to advocacy for safer alternatives including the use of biological control methods. The concern regarding the use of chemical pesticides provides a push for investment in alternative means of control that are environmentally friendly and sustainable, such as biological control.

Developing a biological control for insect pests offers an eco-friendly option to minimize the use of synthetic and environmentally toxic pesticides (Baker et al., 2020; Nawaz et al., 2016). The use of bacteria, protozoa, viruses, nematodes, and fungi in the biological control of insect pests has been widely recognized since the early 19th century (Clausen, 1956; Deka et al., 2021; Nazir et al., 2019). Among these pathogens, fungi are perhaps the most important as they occur naturally, have high virulence, broad host range, and have the least damage to the environment (Deshmukh and Sandhu, 2024a; Mantzoukas and Eliopoulos, 2020). Presently, more than 700 fungal species are regarded as entomopathogenic agents against various arthropod insects (Kidanu and Hagos, 2020; Kumar et al., 2024; Nong et al., 2021). The most common entomopathogenic fungi include *Aspergillus, Entomophthora*, *Isaria, Metarhizium*, *Entomophaga*, and *Beauveria* (Deshmukh and Sandhu, 2024b; Farhan and Kanwal, 2023). The entomopathogenic *Beauveria bassiana* (Hypocreales: Clavicipitaceae) fungi have a wide range of natural hosts, such as Coleoptera, Lepidoptera, Orthoptera, and Diptera (Awan et al., 2021b). This prompted their use in the control of several insect pests (Rajula et al., 2021; Srivani and Jalaja, 2022). *Beauveria bassiana* occurs in soils (Yasin et al., 2019), crops (Khashaba, 2021), and dead insects (Sarker et al., 2024; Wakil et al., 2021), resulting in a variety of local isolates (Keçili et al., 2022). The control of insect pests based on local isolates of *B. bassiana* was deemed more successful and more environmentally suitable (Fite et al., 2020; Rhoda, 2023; Tadele and Emana, 2017).

Acridid insect populations are capabile of thermoregulating their body temperature across different environmental conditions (Blanford et al., 1998; Ouedraogo et al., 2004; Sangbaramou et al., 2018). These insects change thier body temperature when infected with fungi (behavioral fevers). This is achived by assuming basking postures that elevate their internal body temperature above optimal temperatures for microbial growth. As a result, lower mortalities have been reported in field conditions comparared to laboratory conditions when *B. bassiana* formulations are used (Clancy et al., 2018). It has been suggested that the formulation using various carier materials which have higher stability may reduce this behaviour (Lomer et al., 1999; Wakil et al., 2022). Formulating biopesticides entail adding certain active (functional) and non-active (inert) substances to the active ingredients (Bharti and Ibrahim, 2020; Kala et al., 2020). Formulation of indigenous isolates of *B. bassiana* has been found to enhance the efficacy of the isolates against targeted insects from the same environment due to their adaptation to the local conditions (Bayman et al., 2021a). However, there are currently few studies that have tested the efficacy of fungal entomopathogens against desert locust invasions. Therefore the current study aimed at isolating and formulating local isolates of *B. bassiana* and testing their bio-efficacy against adults and 3rd larval instars of desert locusts in Kenya.

2. Materials and methods

2.1. Experimental desert locust and facility

The 21-day experiment was carried out under controlled conditions at the University of Eldoret, at the Department of Biological Sciences. A total of 2400 nymphs and 2350 third stage instars of locusts were purchased from Nairobi University-Chiromo Campus, Department of Entomology laboratory. The locust were transferred to cages in a greenhouse and kept in groups of 30 in cages (45 × 38 × 38 cm H × L × W) made of aluminium with wire mesh sides. The temperature of the greenhouse was maintained at 34 ± 1.2°C and relative humidity at 80 ± 5.5 %. A natural 12 h/12 h light/dark photoperiod was maintained. The desert locusts were fed on wheat grass *adlibitum*.

2.2. Soil sample collection and sites

Soil samples were collected from two agro-ecological zones in Isiolo and Laikipia Counties in Kenya (Table 1). Sampling fields were selected in a systematic manner at 10–15 km interval (Marley *et al*., 2001), resulting in 24 fields in in the two counties. Soils were sampled in three randomly selected plots within the larger field. Soil corers were used to collect soils from a depth of 10–15 cm of the organic and/or the horizon soil zone (Inglis *et al*., 2012). The soil were pooled and mixed, out of which 1 kg sub-sample was taken for further analysis (Idris *et al*., 2007).

To map the sampling points, a GPS receiver was used to get the geocodes and for resampling if need be. Soil samples for isolation of entomopathogens was collected from the grasslands, agricultural farms, forests and controlled areas like national parks that fall on the study field. All samples were taken to University of Eldoret Microbiology Laboratory. Once in the laboratory the soil was preserved at temperature of 22–24 °C and relative humidity of 70–75 %.

2.3. Acquisition of Galleria mellonella larvae

The fifth larval stage of *Galleria mellonella* (Lepidoptera, Pyralidae) were obtained from Dudutek Company in Kenya. The *Galleria mellonella* were kept in the laboratory for 12–24 h and then transferred to different moistened soils obtained from Northern Kenya.

2.4. Isolation and culture of Beauveria bassiana from the soils

Isolation of *B. bassiana* from the soil samples was achieved using *Galleria* bait method (Meyling and Eilenberg, 2006). To heat shock, the *G. mellonella* larvae were immersed in warm water (56 °C) for 5 sec and cooled in cold water (15 °C) for 30 s, before drying in a sterile tissue paper for 1 h. Water was systematically added to 1 kg of the soil sample in 500 ml beaker and the larvae transferred to the moist soil.

Each 500 ml beaker holding ten *G. mellonella* larvae was kept at 25 ± 5°C in the dark. After every 2–3 days the glass containers were inverted and the moisture level checked. The beaker were inspected after every 2 days to remove any dead larvae. The dead larvae were surface sterilized using 1 % sodium hypochlorite for 2–3 min, rinsed in sterile distilled water severally.

The dead larval were transferred to a plate over layered with Whatman filter paper No.1 (Macherey-Nagel, Duren, Germany) moistened with sterile water. Incubation was carried out at 25 ± 2°C in the dark. The sporulated fungi were sub-cultured on Sabourand Dextrose Agar Yeast (SDAY) (Merck Ltd., Darmstadt, Germany) to isolate pure cultures and preserve on SDAY slants at 4 °C.

2.5. Cultivation and maintenance of Beauveria bassiana

Spores of *B. bassiana* were transferred to SDAY plates in 9 cm Petri dishes and incubated at 30 °C for 21 days in the dark. These were inoculated into 50 mL of Sabouraud Dextrose Broth (SDB) liquid broth in a 250 ml Erlenmeyer flask. The mixture was then aerated at 200 rpm on a rotary shaker for 2–3 days at 25 ± 2°C. One ml was pippeted from the flasks on daily basis to monitor the developmental stage and the spore concentration using a Neubauer haemocytometer.

The mycelium was homogenized and inoculated in a blender. A 10 ml of the cultured sample was transferred onto plates containing SDAY and incubated for 14 days at 25 °C. The conidia in the flask were suspended in 15 mL of a solution containing 0.2 % Tween 20 and 0.89 % NaCl. Conidia were harvested by filtrating the suspension using a Buchner-type funnel through a 2-layered filter paper (90 mm). Filtration was done to remove mycelial fragments and aggregated conidia. The resulting mycelilia mat were dried and the spore concentration determined using a Neubauer hemocytometer under a compound microscope (×40).

Single spore fungal cultures were sub-cultured on to several plates of SDAY media and incubated for 10 days. Stock cultures of each isolate were maintained on SDAY slants bottles under refrigeration (4–5 °C) until ready for use.

2.6. Mass production of Beauveria bassiana isolates

Mass production of *B. bassiana* was conducted following a modified method (Mwamburi, 2016). A total of 200 g of rice was weighed and washed four times with sterile distilled water, pre-cooked at 90–100 °C for 15 min and placed in a zipper lock polythene bags and autoclaved for 15 min at 121 °C. They were allowed to cool to room temperature. Exactly 1 ml of conidial isolates were transferred to the polythene bag containing the rice substrate and incubated at 25 °C for 12 h. The bags were manually shaken and incubated for further 1 week at room temperature. Thereafter, the bags were shaken slightly to enhance aeration and incubated for further 2 weeks. The final substrates containing conidia were transferred into plastic basins measuring 4 cm in diameter and 4.5 cm in height and allowed to dry in desiccators by using silica gel for 12 h. The formulations were stored in sealed paper bags in a refrigerator at 4 °C for 2 months. During storage, the moisture content was maintained at 10–30 % and pH 4–7.

2.7. Molecular characterization of putative Beauveria bassiana isolates

2.7.1. DNA extraction

Fungal cultures grown overnight in media were centrifuged at 10,000*g* for 3 min in a falcon tube centrifuge to collect the cells. The supernatant was discarded and the cells resuspended in 200 µl of molecular grade water. DNA was then extracted using the Zymo Bacterial/

Table 1
Soil sample collection sites, habitat and geographical location.

No.	Sample site	Habitat	Soil codes	No. soils samples	Collection region	Altitude	Latitudes and longitude
1	Meteorological station	Weather station	323	5	Isiolo	1109 m	N00°21′14.36″, E037°35′14.40″
2	Shambani scheme	Farms	324	5	Isiolo	1085 m	N00°22′18.10″, E037°34′00.91″
3	Shambani scheme	Open field	326	5	Isiolo	1068 m	N00°22′54.91″, E037°34′17.07″
4	Shambani scheme	Farm	327	5	Isiolo	1060 m	N00°22′58.34″, E037°34′17.28″
5	Shambani primary school	Open field	328	5	Isiolo	1067 m	N00°22′51.68″, E037°34′24.35″
6	Kilimani	Cultivated farm	329	5	Isiolo	1118 m	N00°21′21.45″, E037°33′59.96″
7	Kilimani Farm	Open field	231	5	Isiolo	1115 m	N00°21′15.50″, E037°33′39.89″
8	Kambi Sheikh	Open field	333	5	Isiolo	1174 m	N00°19′48.91″, E037°33′10.59″
9	Buffalo Spring National Park	Park entrance	334	5	Isiolo	865 m	N00°36′24.66″, E037°40′11.97″
10	School of Military Engineering	Open field	335	5	Isiolo	874 m	N00°36′15.02″, E037°40′17.67″
11	Ngenia	Farm	337	5	Lakipia	2114 m	N00°04′35.37″, E037°11′51.45″
12	Mukongodo	Farm	340	5	Lakipia	2130 m	N00°04′28.88″, E037°11′20.55″
13	Chumvi Location	Open field	349	5	Lakipia	2129 m	N00°09′58.06″, E037°12′39.39″
14	Lotach	Forest	350	5	Lakipia	2124 m	N00°09′58.06″, E037°12′39.20″
15	Lotach	Farms	351	5	Lakipia	2135 m	N00°10′19.37″, E037°12′41.70″
16	Natepo	Field	352	5	Lakipia	2136 m	N00°10′07.61″, E037°12′53.21″
17	Gitugi	Forest	338	5	Lakipia	2109 m	N00°04′17.44″, E037°11′04.90″
18	Gitugi	Farm	339	5	Lakipia	2109 m	N00°04′12.83″, E037°11′07.51″
19	Ngenia	Field	341	5	Lakipia	2153 m	N00°04′07.75″, E037°12′13.80″
20	Ministry of Agriculture	Field	336	5	Lakipia	1948 m	N00°00′41.83″, E037°04′38.87″
21	Natepo	Forest	353	5	Lakipia	2133 m	N00°04′04.64″, E037°12′54.32″
22	Ngenia Gulley	Forest	354	5	Lakipia	2130 m	N00°09′53.01″, E037°12′53.76″
23	Ngenia Centre	Open field	355	5	Lakipia	2139 m	N00°09′46.05″, E037°12′54.18″
24	Ngenia Dry place	Open field	356	5	Lakipia	2100 m	N00°09′13.56″, E037°12′53.52″

Fungi DNA Miniprep extraction kit (Zymo Inc, USA) following the manufacturer's instructions. The extracted DNA was quantified and checked for purity (A260/280 ratio) using a Nanodrop 2000C spectrophotometer (Thermofisher Scientific, USA) to ensure that the samples had a minimum concentration (10 ng/μl) and an acceptable quality score (260/280 absorbance ratio).

2.7.2. PCR amplification

PCR amplification of the ITS region was done using the following components: Quickload Master Mix (New England Biolabs, USA), 25 mM Magnesium chloride primers ITS1F(5′-CTTGGTCATTTA-GAGGAAGTAA-3′) and ITS-4R(5′-CCTCCGCTTATTGATATGC-3′), molecular grade water and the DNA template, which resulted to a volume of 20 μl. 35 cycles of PCR amplification was in five steps under different conditions as: 1. initial denaturation (at temperature 94 °C for 5 mins)., 2. Denaturation (at temp. 94 °C for 30 secs), 3. Annealing (temp. 50 °C for 1 mins), 4. Extension (temp. 72 °C for 2 mins) and 5. final extension (temp. 72 °C for 10 mins).

The PCR products were run on a 2 % agarose gel stained with SafeView Classic nucleic acid staining dye (Applied Biological Material, Canada). A 100 bp ladder (New England Biolabs, USA) was used to estimate size of amplified sequence. Electrophoresis was done using sodium borate buffer at 80 V voltage, 400 mA current for 1 h 15 mins. The gel was then visualized under UV light in a transilluminator (Wilber Lourmat XBOX5) to confirm successful PCR amplification.

2.7.3. Sequencing and data processing

The PCR products were sent to Inqaba Biotec, South Africa for Sanger sequencing. The amplicons were cleaned using magnetic beads and sequenced on the ABI3730XL Sanger sequencer platform (Applied Biosystems, USA). The abi format sequences obtained were then visualized, trimmed and converted to FASTA format using the MEGA version 11 software (Tamura *et al.*, 2021).

2.7.4. Species identification and phylogenetics

The FASTA sequences obtained after trimming the Sanger abi sequences were used for species identification using the Basic Local Alignment Search Tool (BLAST) on the National Centre for Biotechnology Information website. The nucleotide collection database was used to search for highly similar sequences (Megablast). Species identification was based on the hit with the highest total score, query coverage and percent identity. For phylogenetics, sequences from the top hits from BLAST results were downloaded as FASTA files and a combined FASTA file created by adding the sequences from the isolates obtained from this study. Multiple sequence alignment of the sequences was done using the CLUSTAL W program implemented on MEGA software, with default parameters and an alignment file created. The alignment file was then used to create a phylogenetic tree on MEGA software using the Maximum Likelihood method using a bootstrap value of1000. A maximum likelihood phylogenetic tree constructed using a bootstrap value of 1000 had 5 clades in total with clade I having most fungal isolates. Six of the seven *Beauveria* isolates obtained from this study clustered in clade I, and one isolate was in clade II. The *B. bassiana* isolates that showed similar accession numbers also displayed some differences in identity.

2.8. Screening of isolated fungi for their biological control potential against locusts

The first screening (10^8 conidia/ml) of the seven confirmed *B. bassiana* isolates was conducted by spraying 1 ml of the aqueous fungal isolates using a 1 L hand sprayer for 5 s, which was timed using a stop watch onto 30 day old locusts. Direct spraying of all locust was ensured through physical examination. The desert locust on cages were provisioned with fresh wheat grass leaves every day and held for observation for 14 days. The study was designed in a completely randomized design with three replicates. Thirty (30) locusts (adults and 3rd nymphal instars) were used in each treatment. The controls experiment was set up containing water and 0.1 % Tween 80. The dead locusts were examined for mycosis. The number of insects that express mycoses were noted. The bioassay was repeated three times. The isolates that caused 90 % screening threshold within a short time were selected for subsequent bioassays.

2.9. Fungal bioassay of entomopathogens against the desert locusts

2.9.1. Formulations of Beauveria bassiana biopesticides

Formulations of the pathogenic strains were done in different carrier materials. These included diatomaceous earth (diatomite/ diatomaceous earth (DE), waste powder milk (whey) and liquid paraffin. The solid carrier materials were dissolved in distilled water and mixed with conidia to make a product with with a concentration of 10^8 conidia/ml. The mixture was serially diluted to make concentration of 10^8, 10^7, 10^6, 10^5 and 10^4 of the substrates prepared using sterile distilled water. During oil formulations, the ingredients were mixed separately at a 50:50 ratios by adding the aqueous phase onto the oil phase to obtain water-in-oil formulation, which was homogenized mechanically using a high-speed homogenizer (Model D-500, Germany). The formulations were diluted with appropriate diluents to the required concentrations.

2.9.2. Determination of stability of formulations

To determine stability of formulations, samples 10 g of solid substrates were taken at after the first month and second months, and serial dilutions 10^{-8}, 10^{-7}, 10^{-6}, 10^{-5} and 10^{-4} of the substrates prepared using sterile distilled water, out of which 1 ml sample were plated onto SDAY plate and incubated. The number of colonies forming units were then determined using colony counter.

2.9.3. Bioassays of desert locusts

Dose response bioassay using different fungal formulations were conducted using a series of conidial concentrations (10^3–10^8 viable conidia/ml) prepared in sterile aqueous 0.01 % Tween 80. A total of 30 adult locusts and 30 nymphs were inoculated by spraying the fungal isolates and formulations. Controls received only the diluents. The experiment was executed in triplicate. The cumulative mortality of each formulation was determined after 14 days. Formulation that showed bioefficiacy at LC90 was used in formulation.

2.10. Data analyses

All data were analyzed using Statistical Package for Social sciences (SPSS version 23.0). The results were presented using graphical techniques such as trend line graphs and bar graphs. Final mortality of adults and 3rd nymphal instars subjected to isolates and formulations of *B. bassiana* were presented as Means ± SEM. Difference in mortality of adults and 3rd nymphal instars were analyzed using One Way ANOVA, and significantly different means discriminated using LSD. The bioassays performed on the desert locust species during the study, consisted of mortality which was considered as response variables against time and spore concentrations of the *B. bassiana* isolates ands formulations. The response variables are binary in nature (died/lived) and naturally follow binomial distribution (Agresti, 1990). Therefore the relationship between the response variables (mortality) and time and concentration (factors) were modeled using Probit analysis (Probit analysis, SPSS). The response frequency was observed as response variables from a total observation of 30 locusts. Meanwhile the days and concentrations were the covariates. To test the strength of the relationships in the Probit plots, slope (Z) statistics was calculated; the lager the Z statistics the larger stronger the isolate or the formulation. The modeled fit was confirmed using chi-square goodness of fit test between the observed response values.

The probits were used to estimate the lethal time to 50 % and 90 %

mortality (LT_{50}) and LT90 respectively; as well as the lethal concentration causing 50 % mortality (LC_{50}) and 90 % mortality (LC_{90}) (Finney, 1971). The LC_{50} and LC_{90} were determined by projecting the Y axis for a probit of 5.00 and 9.00 and taking the inverse $Log_{10}(X)$ of the concentration associated with isolate and the concentrations of the formulations.

The interactions between isolates used in formulation, carrier material and time on the number of colony forming units was determined using generalized linear model, 3-way interactions. Significance of any statistical test was declared at $P < 0.05$.

3. Results

3.1. Bioefficacy of B. bassiana isolates on adults and 3rd nymphal instars

The final mortality (mean% ± SEM) of adult locusts and 3rd nymphal instars subjected to the 7 isolates of *Beauveria bassiana* after 14 days post-incubation are shown Fig. 1. All the tested isolates of *B. bassiana* were significantly better than control ($P < 0.05$) and thus pathogenic to the adults and 3rd nymphal instars of *S. gregaria* under laboratory conditions. They caused mortality ranging from 57.8–100 % after 14 days post-incubation. There were significant differences in mortality of adult *S. gregaria* subjected to the 7 isolates of *B. bassiana* ($F = 410.234$, $df = 7$, $P < 0.001$). The highest mortality in the adults occurred when the insects were treated with isolate 341 (100 %), followed by isolate 231 (96.3 %) and then 334 (89.9 %) while the least efficacious isolates were 349 (mortality = 57.9 %) and 351 (mortality = 57.8 %). Similarly, the mortality of 3rd nymphal instar subjected to the 7 isolates of *Beauveria bassiana* were significant ($F = 176.534$, $df = 7$, $P < 0.001$). Similar trend were observed in the mortality of 3rd nymphal instars subjected to 7 isolates, where the highest mortality occurred when the insect stage were treated with isolates 341 (99.6 %) followed by 231 (both 95.6 %) and then 334 (91.2 %) while the least efficacious isolates were 349 and 351 (mortality = 66.0 to 67.7 %) after 14 days post-incubation.

Daily trends in the mortality of adults and 3rd nymphal instars subjected to the 7 isolates of *Beauveria bassiana* after 14 days post-incubation is shown Fig. 2. Mortality trends of the adults and 3rd nymphal instars were consistently higher in 3 isolates (341, 231 and 334) exhibiting the trend: 341 > 231 > 334 over the entire experimental period.

The LT_{50} and LT_{90} of the 7 isolates of *Beauveria bassiana* on adult and 3rd nymphal instar of *S. gregaria* after 14 days post-incubation are presented in Table 2 while the statistical test of significance of the mortality-time response is shown in Table 3. For adults, the isolates 341, 231, and 334 elicited 50 % mortality responses after 4.80 days, 4.92 days and 5.68 days respectively. Based on the slope of the mortality-time response, isolate 341 was the most effective (Slope [Z] = 11.553), followed by 231 (Slope [Z] = 10.831) and then isolate 334 (Slope [Z] = 10.421). The least effective isolates were 349 (Slope [Z] = 8.384) and 351 (Slope [Z] = 7.667). Meanwhile the isolates 341, 231, and 334 caused 90 % mortality of adult locust (LT_{90}) after 10.68 days, 13.44 days and 15.61 days respectively. They were the only isolate to cause mortality to 90 % of the adult locusts within the 14 days experimental period where isolates 333, 339, 349 and 351 failed.

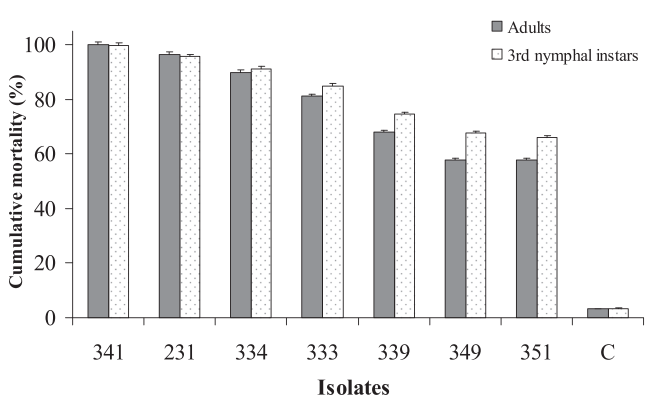

Fig. 1. Final mean mortality (% ± SEM) of adult and 3rd nymphal instar subjected to 7 isolates of *Beauveria bassiana* after 14 days post-incubation. C is control.

In the 3rd nymphal instars, isolates 341, 231, and 334 elicited 50 % mortality response after 4.96 days, 6.26 days and 7.52 days respectively. Slope analysis of the mortality-time response for the 3rd nymphal instars indicated that isolate 341 was the most effective (Slope [Z] = 11.261), followed by 231 (Slope [Z] = 10.792) and then isolate 334 (Slope [Z] = 10.009). The least effective isolates were 349 (Slope [Z] = 7.209) and 351 (Slope [Z] = 6.727). Meanwhile the same isolates' LT_{90} against 3rd star nymphal instars were 10.93 days, 13.98 days and 14.73 days respectively. However, isolates 333, 339, 349 and 351 were not capable of killing 90 % of the 3^{rd} nymphal instars within the 14 days period. All statistical parameters and chi-square tests test of goodness supported the validity of the current results.

Mortality of adults and 3^{rd} nymphal instars subjected to different concentrations *Beauveria bassiana* isolates are presented in Fig. 3. Mortality trends of the adults and 3^{rd} nymphal instars followed dose–response relationship being consistently higher in 3 isolates (341, 231 and 334) exhibiting the trend: 341 > 231 > 334.

The LC_{50} and LC_{90} of the 7 isolates of *Beauveria bassiana* on adult and 3rd nymphal instar of *S. gregaria* subjected to different concentrations of *Beauveria bassiana* isolates are presented in Table 4 with the requisite tests of statistical significance of the mortality-concentration response is shown in Table 5. Isolates 341, 231, and 334 elicited 50 % mortality responses at concentrations 1.1×10^5 conidia/ml, 2.5×10^5 conidia/ml and 1.7×10^6 conidia/ml respectively. Slope analysis of the mortality-concentration response, indicate that isolate 341 elicited response at the least concentration (Slope [Z] = 9.645), followed by 231 (Slope [Z] = 8.555) and then isolate 334 (Slope [Z] = 8.192). The least effective isolates that required higher concentration to elicited effective response were 349 (Slope [Z] = 6.475) and 351 (Slope [Z] = 5.870). Meanwhile the isolates 341, 231, and 334 were lethal to 90 % of adult locust (LC_{90}) at concentration 2.1×10^7 conidia/ml, 6.7×10^7 conidia/ml and 1.6×10^9 conidia/ml respectively. They were the only isolate to cause mortality to 90 % of the adult locusts within the 10^8 conidia/ml in the undiluted isolates.

In the 3rd nymphal instars, isolates 341, 231, and 334 elicited 50 % mortality responses to 50 % (LC_{50}) of the locusts at concentrations 1.1×10^4 conidia/ml, 2.9×10^4 conidia/ml and 5.5×10^6 conidia/ml respectively. The slope of the mortality-concentration response, suggest that isolate 341 caused mortality to the locusts at the least concentration (Slope [Z] = 9.069), followed by 231 (Slope [Z] = 8.863) and then isolate 334 (Slope [Z] = 8.407). The least effective isolates that required higher concentration to elicited effective response were 351 (Slope [Z] = 7.416) and 339 (Slope [Z] = 6.711). Meanwhile the isolates 341, 231, and 334 were lethal to 90 % of 3rd nymphal instars (LC_{90}) at concentration 6.9×10^5 conidia/ml, 2.0×10^6 conidia/ml and 1.4×10^7 conidia/ml respectively. They were the only isolate to cause mortality to 90 % of the adult locusts within the 10^8 conidia/ml in the undiluted isolates.

3.2. Bio-efficacy of B. bassiana formulations

Results showing the mean mortality (% ± SE) of adult and 3rd nymphal instar subjected to formulation of the *Beauveria bassiana* isolates after 14 days post-incubation is shown Fig. 4. In terms of mortality effects, all formulation resulted in mortality in adult and third nymphal instars > 65 %. There were significant differences in mortality of adult and third nymphal instarts of *S. gregaria* subjected to the different formulations of *B. bassiana* ($P < 0.001$). Formulations with 341–1, 341–2,

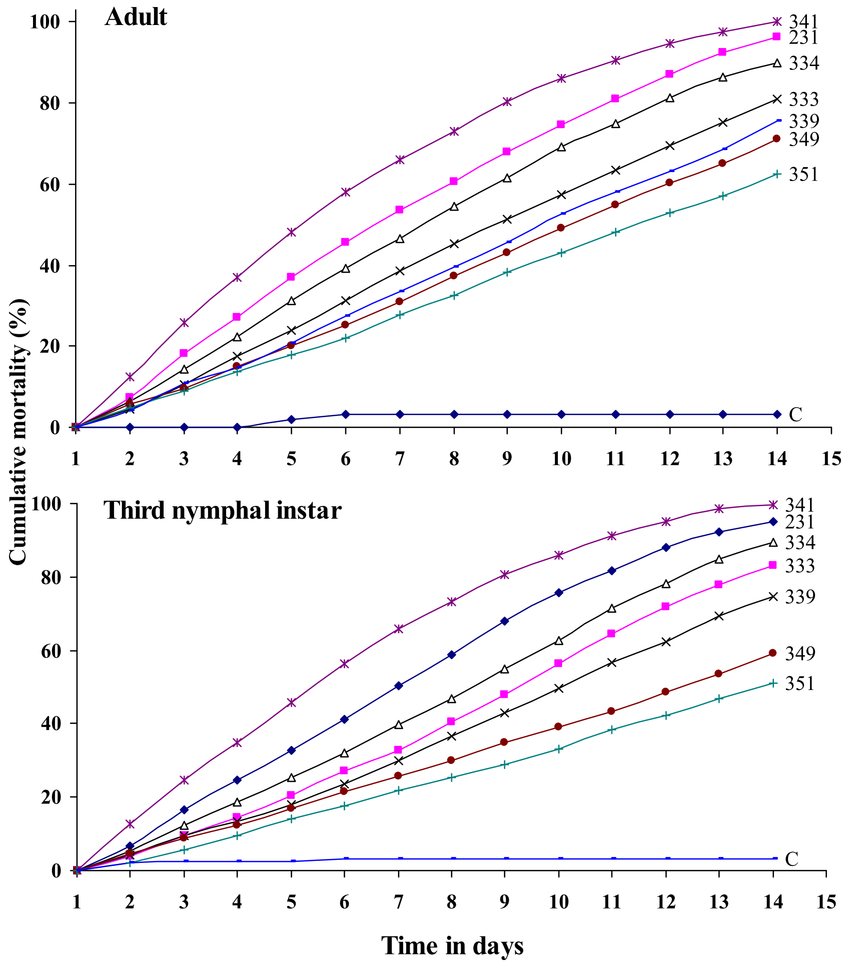

Fig. 2. Daily trends in mortality of adults and 3rd nymphal instars subjected to the *Beavaria bassiana* isolates for the 14 days experiment period. C is negative control.

Table 2
The LT_{50} and LT_{90} of the 7 isolates of *Beauveria bassiana* on adult and 3rd nymphal instar of *S. gregaria* after 14 days post-incubation.

	Mortality parameters estimates			
	Adults		3rd nymphal instars	
Isolates	LT_{50} (±0.95CL)	LT_{90} (±0.95CL)	LT_{50} (±0.95CL)	LT_{90} (±0.95CL)
341	4.80 (4.28–5.29)	10.68 (9.41–12.63)	4.96 (4.45–5.46)	10.93 (9.69–12.76)
231	5.92 (5.35–6.50)	13.44 (11.76–16.06)	6.26 (5.68–6.85)	13.98 (12.23–16.74)
334	6.68 (6.05–7.34)	13.61 (13.46–19.16)	7.52 (6.83–8.26)	14.73 (13.08–22.34)*
333	8.15 (7.36–9.08)	21.14 (17.34–28.26)*	8.50 (7.73–9.41)	20.33 (16.94–26.56)*
339	9.10 (8.16–10.30)	25.19 (19.93–36.42)*	9.65 (8.65–10.97)	26.12 (20.57–37.72)*
349	9.71 (8.63–11.18)	28.82 (22.06–43.72)*	12.56 (10.75–15.78)	44.15 (30.08–85.04)*
351	11.24 (9.80–13.53)	37.01 (25.54–63.97)*	14.47 (12.15–19.07)*	49.12 (32.55–60.42)*

*Not capable of eliciting mortality within the 14 days experimental period.

and 334–1 had the highest efficacy (>99 %) against the adult locusts. Meanwhile in the nymphal instars, the highest mortality was obtained with formulations 341–1, 341–3, 334–1 and 334–2 (>98 % mortality).

Daily trends in the mortality of adults and 3rd nymphal instars subjected to formulation of *Beauveria bassiana* isolates after 14 days post-incubation is shown Fig. 5. Mortality trends of the adults and 3rd nymphal instars were consistently higher in 4 formulations (341–1(Oil), 341–2(DE) and 334–1(Oil).

The LT_{50} and LT_{90} of *Beauveria bassiana* formulations on adult and 3rd nymphal instar of *S. gregaria* after 14 days post-incubation are presented in Table 6 while the statistical test of significance of the mortality-time response is shown in Table 7. From the range of doses used, the mortality effect from exposure observation was found to be time dependant with good model fit (P values < 0.05).

In the adult locusts formulations 341–1, 341–2 and 334–1 elicited 50 % mortality responses after 4.27 days, 4.84 days and 4.85 days respectively. Based on the slope of the mortality-time response, formulation 341–1 was the most effective (Slope [Z] = 11.696), followed by 341–2 (Slope [Z] = 11.240) and then formulation 334–1 (Slope [Z] = 10.633). The least effective formulations were 231–1 (Slope [Z] = 8.039), 231–3 (Slope [Z] = 8.479) and 231–2 (Slope [Z] = 8.584). The formulation 341–1, 341–2 and 334–1 caused 90 % mortality of adult locust (LT_{90}) after 8.10 days, 9.45 days and 10.45 days respectively. Together with formulations 341–3, 334–2 and 334–3 they cause mortality to 90 % of the adult locusts within the 14 days experimental period.

In the 3rd nymphal instars, isolates 341–1, 341–3, 334–1 and 334–2 elicited 50 % mortality responses (LT_{90}) after 4.32 days, 4.51 days, 4.66 days and 4.86 days respectively. This shows that oil based formulation was the best followed by whey and then DE, although the overall outcome was affected by the isolates Slope analysis of the mortality-concentration response for the 3rd nymphal instars indicated that formulation 341–1 was the most effective (Slope [Z] = 11.714),

Table 3
Statistical test of significance for the adults and 3rd stage nymphal instars for the mortality response within the 14 days trial period.

	Probit parameter estimates								Chi-square goodness of fit test			
	Adults				3rd nymphal instar				Adults		3rd nymphal instar	
Isolates	*Par. Est.*	*Std. error*	*Slope (Z)*	*P-value*	*Par. Est.*	*Std. error*	*Slope (Z)*	*P-value*	χ^2	*P*-value	χ^2	*P*-value
341	3.687	0.343	11.553	<0.0001	3.738	0.332	11.261	<0.0001	61.498	<0.0001	16.713	0.0004
231	3.604	0.334	10.831	<0.0001	3.678	0.341	10.792	<0.0001	58.162	<0.0001	56.186	<0.0001
334	3.479	0.334	10.421	<0.0001	3.438	0.342	10.009	<0.0001	59.841	<0.0001	15.984	0.0009
333	3.092	0.332	9.338	<0.0001	3.384	0.357	9.482	<0.0001	48.656	<0.0001	48.452	<0.0001
339	2.898	0.331	8.761	<0.0001	2.962	0.343	8.627	<0.0001	27.461	0.0005	27.462	<0.0001
349	2.713	0.324	8.384	<0.0001	2.349	0.326	7.209	<0.0001	11.172	0.0003	11.117	0.0052
351	2.472	0.323	7.667	<0.0001	2.414	0.359	6.727	<0.0001	3.304	0.0235	11.246	0.0042

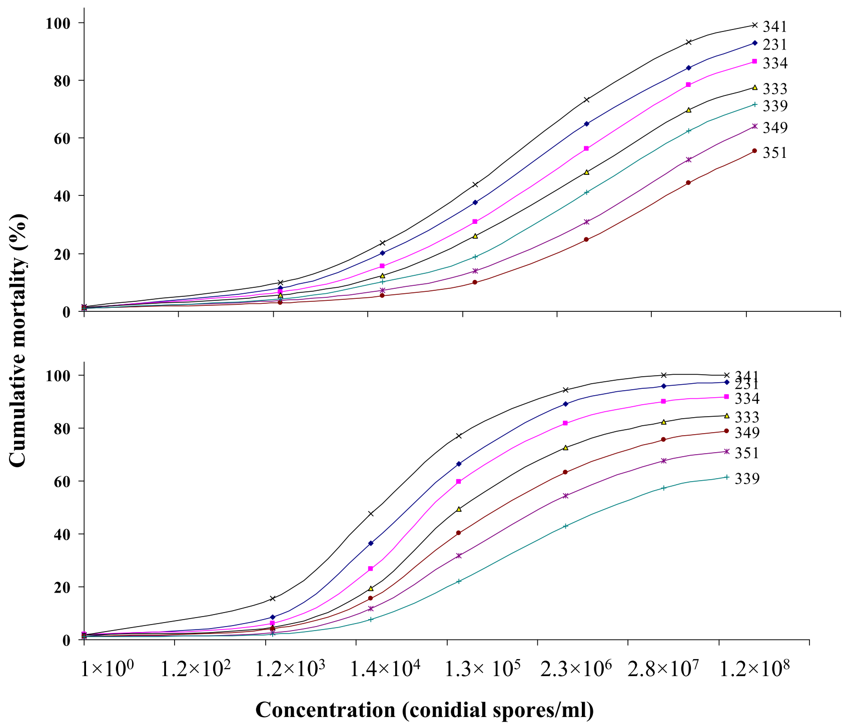

Fig. 3. Mortality of adults and 3rd nymphal instars subjected to different concentration *Beavaria bassiana* subjected to different concentrations of *Beavaria bassiana* isolates.

followed by 341–3 (Slope [Z] = 11.671) and then formulation 334–1 (Slope [Z] = 11.582) and finally formulation 334–2 (Slope [Z] = 11.364). The least effective formulations were 231–3 (Slope [Z] = 8.673), 231–1 (Slope [Z] = 8.442) and 231–2 (Slope [Z] = 8.209). In the meantime the formulation 341–1, 341–3 and 334–1 and 334–2 caused 90 % mortality of adult locust (LT_{90}) after 9.10 days, 10.79 days, 10.98 days and 11.66 days respectively. All statistical parameters and chi-square tests test of goodness supported the validity of the current results.

Mortality of adults and 3rd nymphal instars subjected to different concentrations *Beauveria bassiana* formulations are presented in Fig. 6. Mortality trends of the adults and 3rd nymphal instars followed dose–response relationship being consistently higher in 4 formulations (341–1, 341–2, 341–3 as well as 334–1).

The LC_{50} and LC_{90} of the nine formulations on adult and 3rd nymphal instar subjected to different concentrations of *Beauveria bassiana* formulations are presented in Table 8 and the concurrent tests of statistical significance of the mortality-concentration response is shown in Table 9. Formulations 341–1, 341–2, 341–3 and 334–1 elicited 50 % mortality responses at concentrations 9.2×10^3 conidia/ml, 1.5×10^4 conidia/ml, 3.2×10^4 conidia/ml and 7.5×10^4 conidia/ml respectively. Slope analysis of the mortality-concentration response, indicate that formulations 341–1 elicited response at the least concentration (Slope [Z] = 9.830), followed by 341–2 (Slope [Z] = 9.026), then 341–3 (Slope [Z] = 8.437) and finally 334–1 (Slope [Z] = 8.260). The least effective isolates that required higher concentration to elicited effective response were 231–1 (Slope [Z] = 7.798), 231–1 (Slope [Z] = 7.295) and 231–2 (Slope [Z] = 6.744). Meanwhile the isolates 341–1, 341–2, 341–3 and 334–1 were lethal to 90 % of adult locust (LC_{90}) at concentration 4.7×10^5 conidia/ml, 8.6×10^6 conidia/ml, 2.2×10^6 conidia/ml and 1.2×10^7 conidia/ml respectively.

In the 3rd nymphal instars, Formulations 341–1, 341–2, 341–2 and 334–1 elicited 50 % mortality responses at concentrations 1.1×10^5 conidia/ml, 1.1×10^5 conidia/ml, 2.5×10^5 conidia/ml and 1.7×10^6 conidia/ml respectively. Slope analysis of the mortality-concentration response, indicate that formulations 341–1 elicited response at the least concentration (Slope [Z] = 9.645), followed by 341–2 (Slope [Z] = 8.555), then 341–2 (Slope [Z] = 9.645) and finally 334 (Slope [Z] = 8.192). The least effective isolates that required higher concentration to elicited effective response were 231–3 (Slope [Z] = 8.673), 231–1 (Slope [Z] = 8.442) and 231–2 (Slope [Z] = 6.475). Meanwhile the isolates 341–1, 341–2, 341–2 and 334–1 were lethal to 90 % of adult locust (LC_{90}) at concentration 2.1×10^7 conidia/ml, 2.1×10^7 conidia/ml, 6.7×10^7 conidia/ml and 1.6×10^9 conidia/ml respectively.

Table 4
The LC_{50} and LC_{90} of the 7 isolates of *Beauveria bassiana* on adult and 3rd nymphal instar of *S. gregaria* subjected to different concentration *Beauveria bassiana* isolates.

Isolates	Mortality parameters			
	Adults		3rd nymphal instars	
	LC_{50} (±0.95CL)	LC_{90} (±0.95CL)	LC_{50} (±0.95CL)	LC_{90} (±0.95CL)
341	1.1×10^5 (4.6×10^4–3.1×10^5)	2.1×10^7 (5.3×10^6–1.8×10^8)	1.1×10^4 (4.9×10^3–2.6×10^4)	6.9×10^5 (2.3×10^5–3.9×10^6)
231	2.5×10^5 (9.7×10^4–6.4×10^5)	6.7×10^7 (1.7×10^6–4.7×10^8)	2.9×10^4 (1.2×10^4–6.5×10^5)	2.0×10^6 (7.2×10^5–8.7×10^6)
334	1.7×10^6 (6.0×10^5–6.0×10^6)	1.6×10^9 (2.4×10^8–3.4×10^{10})	5.5×10^4 (2.9×10^4–1.8×10^5)	1.4×10^7 (4.3×10^6–7.6×10^7)
333	6.2×10^5 (2.3×10^5–1.7×10^6)	2.9×10^9 (6.1×10^8–3.0×10^9) *	2.3×10^5 (8.4×10^4–6.1×10^5)	9.4×10^7 (2.2×10^7–7.4×10^9)*
339	4.7×10^6 (1.4×10^6–2.0×10^7)	6.5×10^9 (7.2×10^8–2.5×10^{11})*	8.8×10^6 (2.5×10^6–4.7×10^7)	2.1×10^{10} (1.6×10^9–1.7×10^{12})*
349	1.8×10^7 (4.7×10^6–1.2×10^8)	4.8×10^{10} (3.1×10^9–6.5×10^{12})*	6.1×10^5 (2.1×10^5–1.8×10^6)	4.1×10^8 (8.0×10^7–5.1×10^9)*
351	6.8×10^7 (1.3×10^7–9.2×10^8)	3.4×10^{11} (1.9×10^{10}–2.3×10^{13}) *	1.9×10^6 (6.3×10^5–6.7×10^6)	2.2×10^9 (3.1×10^8–5.4×10^{10})*

*Did not cause mortality at the 10^8 conidia/ml in the undiluted isolates. Concentration above 10^8 were based on projections from the probit model.

3.3. Stability of different concentration of B. bassiana isolates

The quantitative counts of *B. bassiana* Cfu in different formulations following dilutions and formulation with different carrier materials are provided in Table 10 while the tests of between-subjects effects between the treatments and stability (cfu) measures at 1 and 2 months are shown in Table 11. The main effect of treatments (isolates), carrier materials and time significantly influenced the Cfu. Generally, increase in Cfu was highest in formulation with isolate 341 regardless of the carrier material and time. Also Cfu was higher after 1 month compared to 2 months. The carrier material significantly affected the Cfu where Whey was the best after 1 month and liquid paraffin was best after 2 months. There was a significant 3-way interaction ($P < 0.05$) of isolate for formulation, carrier material and time in determining the Cfu of the *B. bassiana* formulations. After 1 month, the best Cfu occurred in formulation with isolates 231 and 341 formulated using DE, while highest Cfu was observed in formulation with isolate 334 formulated with either liquid paraffin or whey. After 2 months, the highest Cfu occurred in formulation with isolate 231, 341 and 334 formulated using liquid paraffin.

4. Discussion

Beauveria bassiana has been the focus of commercial-scale efforts to use fungi to manage insect pests in the past 20–30 years. This has been achieved through isolation of *B. bassiana* from several environmental media. In the current study, seven *B. Bassiana* isolates were found in the soils taken from 24 different local areas situated in Laikipia and Isiolo counties of Kenya where there have been cyclic outbreaks of desert locust in the past. Presence of *B. bassiana* in the environment may be influenced by several factors such as site (Wang et al., 2022) soil organic matter content (Sharma et al., 2021), soil biotic assemblage (Majchrowska-Safaryan and Tkaczuk, 2021), and the weather or climatic condition of the region (Reyes-Haro et al., 2024). This study used two stage screening to isolates and identify virulent *B. bassiana* strains which has been used effectively by previous workers (Camara et al., 2022; Qiu et al., 2021). In our study 18 isolates were screened based on morphological parameters from where seven were selected based on molecular technique. From the seven isolates, the best three isolates were selected based on pathogenicity assays *in vivo* in which isolates caused 90 % mortality (LC_{90}) in concentrations less than the conidial concentration of the original isolates (10^8 conidia/ml). Via this approach the three *B. bassiana* isolates (341, 231 and 334), which showed the lowest LC_{90} at 10^5 conidia/ml were used for making biopesticide formulation using three carrier materials: liquid paraffin, Diatomaceous earth and whey

All the tested isolates of *B. bassiana* were significantly better than control ($P < 0.05$) and thus pathogenic to the adults and 3rd nymphal instars of *S. gregaria* under laboratory conditions. Following exposure of adults and 3rd nypmphal instars, to the isolates, it was observed that *S. gregaria* became slowly inactive within 8–24 h, had slow feeding and on the 3rd and 4th day, the abdomen of the adult locust becomes red, following by whitish colonization by *B. bassiana*. The colony spread to the entire locust leading to death of the locust. Therefore, it was evident that all the seven tested isolates of *B. bassiana* were pathogenic to the adults and 3rd nymphal instars of *S. gregaria*, causing mortality ranging from 63–100 %. *Beauveria bassiana* possess the potential to produce infections conidia, which penetrate the insect's cuticle, indicating that,

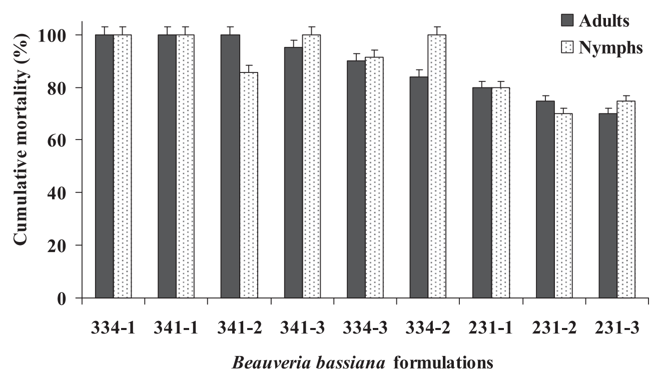

Fig. 4. Mean mortality (% ± SEM) of adult and 3rd nymphal instar subjected to different formulations of *Beauveria bassiana* isolates after 14 days post-incubation. Carrier materials: 1 = Oil, 2 DE and 3 = Whey.

Table 5
Statistical test of significance for the adults and 3rd stage nymphal instars for the mortality response under different concentrations of *Beauveria bassiana* isolates.

Isolates	Probit parameter estimates								Chi-square goodness of fit test			
	Adults				3rd nymphal instars				Adults		3rd nymphal instar	
	Par. Est.	*Std. error*	*Slope (Z)*	*P-value*	*Par. Est*	*Std. error*	*Slope (Z)*	*P-value*	χ^2	*P*-value	χ^2	*P*-value
341	0.566	0.074	9.645	<0.0001	0.724	0.102	9.069	<0.0001	37.173	0.0093	25.211	0.0006
231	0.528	0.062	8.555	<0.0001	0.699	0.084	8.863	<0.0001	27.475	0.0097	40.490	<0.0001
333	0.431	0.057	7.602	<0.0001	0.491	0.058	8.473	<0.0001	20.831	0.0010	42.382	<0.0002
334	0.480	0.059	8.192	<0.0001	0.563	0.064	8.407	<0.0001	28.172	0.0092	46.753	0.0012
339	0.408	0.057	7.13	<0.0001	0.381	0.057	6.711	<0.0001	26.931	0.0095	25.598	<0.0001
349	0.374	0.058	6.475	<0.0001	0.453	0.056	8.024	<0.0001	31.323	0.0003	30.895	0.0012
351	0.306	0.059	5.870	<0.0001	0.417	0.056	7.416	<0.0001	33.634	0.0023	29.293	0.0032

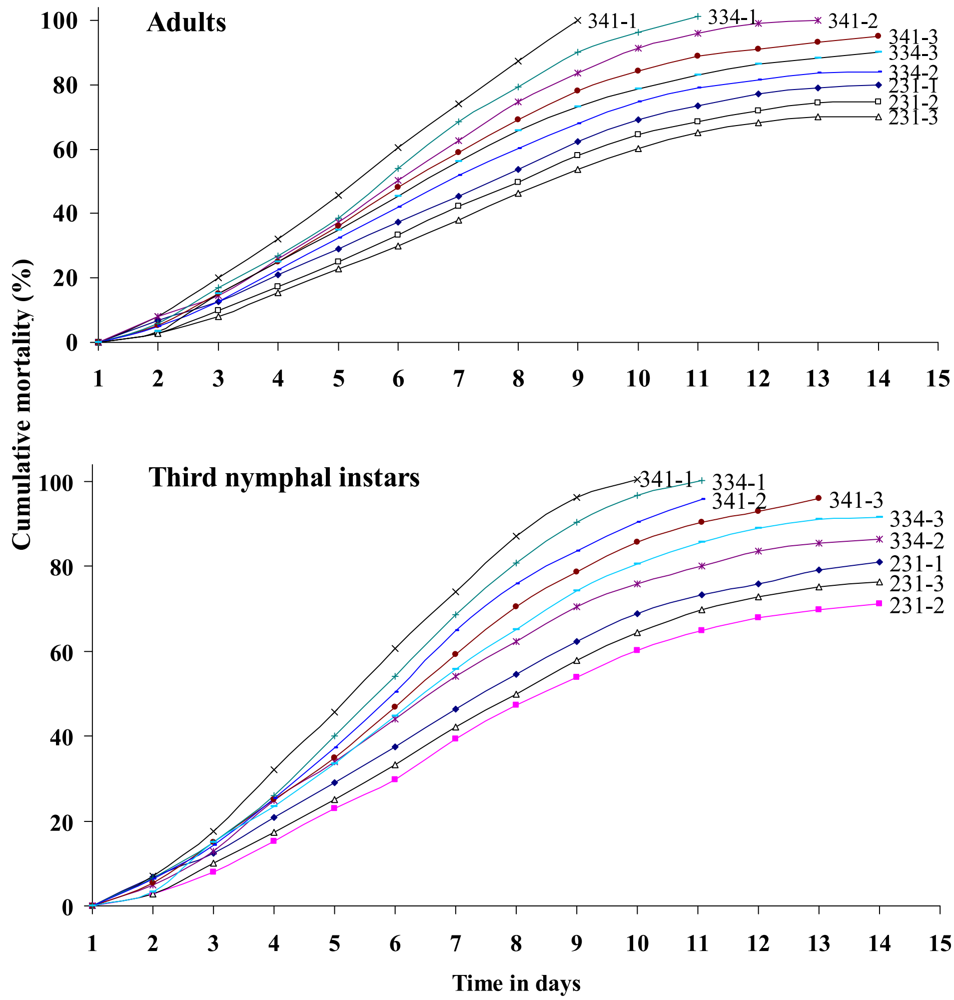

Fig. 5. Daily trends in mortality of adult locusts and 3rd nymphal instars in different formulations of isolates after 14 days post-incubation. Carrier materials: 1 = Oil, 2 DE and 3 = Whey.

Table 6
The LT_{50} and LT_{90} of the *Beauveria bassiana* formulations on adult and 3rd nymphal instar of *S. gregaria* after 14 days post-incubation.

	Mortality parameters			
	Adults		3rd nymphal instars	
Formulations	LT_{50} (±0.95CL)	LT_{90} (±0.95CL)	LT_{50} (±0.95CL)	LT_{90} (±0.95CL)
341-1	4.27 (3.47–5.01)	8.10 (6.79–10.58)	4.32 (3.46–5.12)	9.10 (6.72–10.84)
341-2	4.84 (3.96–5.69)	9.45 (8.11–12.45)	5.74 (5.12–8.35)	14.55 (12.50–17.85)
334-1	4.85 (4.32–5.43)	10.45 (9.11–13.25)	4.66 (3.72–5.54)	10.98 (7.40–12.22)
341-3	5.18 (4.66–5.69)	11.33 (10.05–13.23)	4.51 (4.34–5.90)	10.79 (9.11–13.85)
334-3	5.53 (4.95–6.09)	13.17 (11.48–18.79)	5.48 (4.93–6.03)	12.48 (10.97–14.71)
334-2	5.60 (4.71–6.56)	12.55 (10.30–17.16)	4.86 (3.95–5.72)	11.66 (9.00–12.95)
231-1	6.48 (5.76—7.24)	18.62 (15.38–24.41)	6.46 (5.75–7.23)	18.75 (15.35–24.33)
231-2	7.22 (6.44–8.09)	20.99 (17.05–28.35)	7.79 (6.94–8.80)	23.68 (18.81–33.32)
231-3	7.84 (6.98–5.65)	23.90 (18.94–33.73)	7.15 (6.39–8.00)	20.44 (16.70–27.34)

B. bassiana induces an appropriate mechanism to overcome the insect's cellular defense system. The isolates 341, 231, and 334 caused 50 % mortality responses after 4.80 days, 4.92 days and 5.68 days respectively and were the best among the 7 isolates. The percent mortality levels from different studies are not comparable precisely because of various factors such as different strains and concentrations, diverse ecological situation, and local test insect culture. The current study are in agreement with entomopathogenicity of different fungal isolates against lace bug, *Corythucha arcuata* coffee berry borer, *Hypothenemus hampei* (Ferrari) (Bayman et al., 2021b), greater wax moth, Galleria mellonella larvae (Habtegebriel et al., 2016; Safavi, 2013). Similar results were also reported for *B. bassiana* against the 1st, 2nd and 3rd instar larvae of the jute hairy caterpillar, *Spilarctia oblique* (Bhadauria et al., 2013), immature stages of fall army worm, *Spodoptera frugiperda* (Smith) (Idrees et al., 2022) and bronze bug, *Haumastocoris peregrinus* (Santos et al., 2018) as well as the red palm weevil (RPW), *Rhynchophorus ferrugineus* (Yasin et al., 2019). The highest mortality in adults and 3rd nymphal instar occurred when the insects were treated with isolate 341, 231 and 334 which appeared to be better at controlling desert locust than other isolates. Virulence of different isolates may be increased in areas where they had contacts with the target species (Yassin *et al.*, 2017). It is also possible to suggest that differences in the level of virulence of isolates may be due to microbe's selection, recombination and mutation as isolates were collected from diversified ecological regions which might have influenced their genetic make-up.

Evidence that the fungal isolates have a good level of efficacy under

Table 7
Statistical test of significance for the adults and 3rd stage nymphal instars in *B. bassiana* formulations after 14 days post-incubation.

	Probit parameter estimates								Chi-square goodness of fit test			
	Adults				3rd nymphal instar				Adults		3rd nymphal instar	
Isolates	*Par. est*	*Std. error*	*Slope (Z)*	*P-value*	*Par. Est*	*Std. error*	*Slope (Z)*	*P-value*	χ^2	*P*-value	χ^2	*P*-value
341-1	4.608	0.394	11.696	<0.0001	4.696	0.401	11.714	<0.0001	33.034	<0.0001	38.543	<0.0001
341-2	4.217	0.356	11.240	<0.0001	3.172	0.296	10.721	<0.0001	32.432	<0.0001	45.690	<0.0001
334-1	3.647	0.431	10.633	<0.0001	4.498	0.378	11.582	<0.0001	19.471	0.0021	40.224	<0.0001
341-3	3.773	0.328	10.520	<0.0001	3.974	0.341	11.671	<0.0001	14.854	0.0025	23.633	<0.0001
334-3	3.399	0.308	11.046	<0.0001	3.591	0.319	11.067	<0.0001	16.798	0.0006	20.112	0.0004
334-2	3.691	0.326	11.307	<0.0001	4.301	0.363	11.364	<0.0001	26.762	0.0001	35.172	<0.0001
231-1	2.796	0.281	8.039	<0.0001	2.797	0.281	8.442	<0.0001	38.125	<0.0001	35.219	<0.0001
231-2	2.764	0.289	8.479	<0.0001	2.656	0.288	8.209	<0.0001	38.411	<0.0001	37.852	<0.0001
231-3	2.648	0.288	8.584	<0.0001	2.811	0.291	8.673	<0.0001	39.522	<0.0001	43.432	<0.0001

Carrier materials: 1 = Oil, 2 = DE and 3 = Whey.

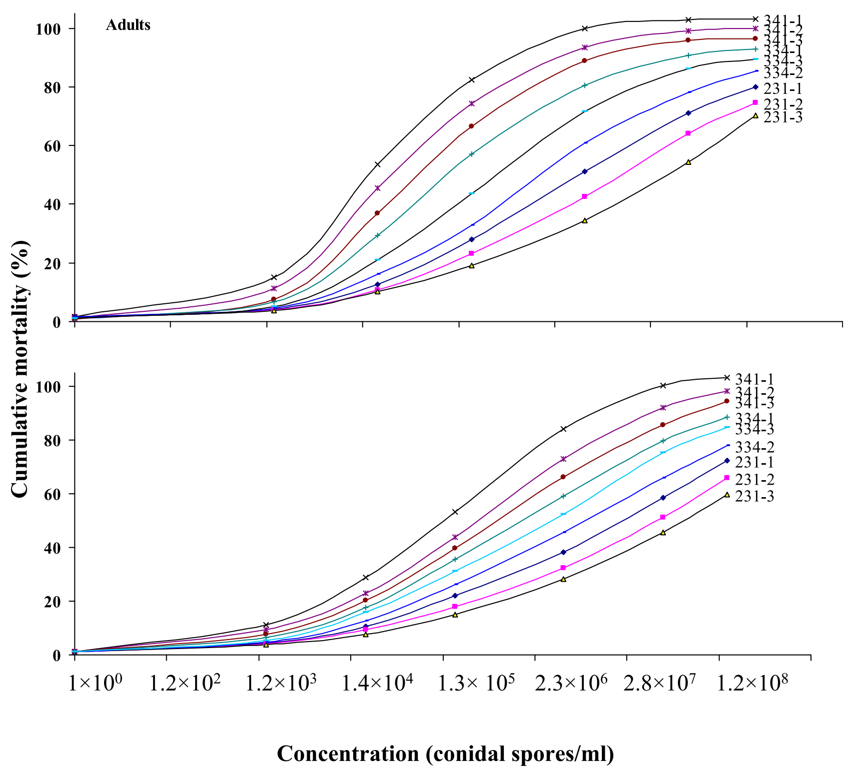

Fig. 6. Mortality of adults and 3rd nymphal instars subjected to different concentration *Beavaria bassiana* isolate.

different concentrations is essential if biopesticides based on *B. bassiana* are to be considered. In the current study, mortality of adults and 3rd nymphal instars subjected to different concentrations *Beauveria bassiana isolates* showed that isolates 341, 231, and 334 elicited 50 % mortality (LC_{50}) at concentrations 1.1×10^4 conidia/ml, 2.9×10^4 conidia/ml and 5.5×10^6 conidia/ml respectively. It is well known that fungal disease in an insect pest population are dosage-dependent (Pu et al., 2023) including in *B. bassiana* (Biryol et al., 2021). The current study are in agreements with previous study on the pathogenicity of *B. bassiana* to cowpea aphid, *Aphis craccivora* (Akrich et al., 2023), third instar larvae of cotton ballworm, *Helicoverpa armigera* (Bisandre et al., 2023), adult spotted-wing drosophila, *Drosophila suzukii* (Gutierrez-Palomares et al., 2021) and adults of the cowpea weevil, *Callosobruchus maculatus* (Ozdemir et al., 2020).

Formulation may enhance the infectivity of the fungal conidia and allow a product to be stored over a prolonged period, it also leads to stability and ease the delivery, protect the agent from environmental influence at the target site, as well as enhance interaction between the agent(s) and the target pest(s) (Behle and Birthisel, 2023; Kumawat et al., 2014). In the current study, all formulation resulted in mortality in adult and third nymphal instars > 65 %. Formulations with 341–1, 341–2, and 334–1 had the highest efficacy (>99 %) against the adult locusts. Meanwhile in the nymphal instars, the highest mortality was obtained with formulations 341–1, 341–3, 334–1 and 334–2 (>98 % mortality). These results suggest that oil based formulation was the best followed by DE although the overall outcome was affected by the isolates. The efficacy of the oil formulations of two *B. bassiana* strains were shown to be effective in controlling cereal aphids (Qi et al., 2024), greater grain weevil, *Sitophilus zeamais* (Motschulsky) (Hidalgo et al., 1998). It has been postulated that oil formulations were better enhances fungal virulence toward insect. Since insect cuticle especially epicuticle (lipid layer) is the primary site of establishment of mycosis, oil formulation increase the adhesion of spore to the insect cuticle through hydrophobic interaction between the spore and cuticle surface (Awan et al., 2021a; Wraight et al., 2016).

Mortality trends of the adults and 3rd nymphal instars followed dose–response relationship being consistently higher in 4 formulations 341–1, 341–2, 341–3 and 334–1 which elicited 50 % mortality responses

Table 8
The LT_{50} and LT_{90} of the 7 isolates of *Beauveria bassiana* on adult and 3rd nymphal instar of *S. gregaria* after 14 days post-incubation.

	Mortality parameters			
	Adults		3rd nymphal instars	
Formulations	LC_{50} (±CI)	LC_{90} (±CI)	LC_{50} (±CI)	LC_{90} (±CI)
341-1	9.2×10^3 (8.8×10^3–2.3×10^4)	4.7×10^5 (1.3×10^5–4.8×10^6)	6.2×10^4 (2.2×10^4–1.8×10^5)	9.2×10^4 (1.9×10^6–1.4×10^8)
341-2	1.5×10^4 (6.8×10^3–3.5×10^4)	8.6×10^6 (2.8×10^5–4.8×10^6)	1.2×10^5 (5.1×10^4–3.2×10^5)	2.4×10^7 (5.9×10^6–2.1×10^8)
341-3	3.2×10^4 (1.3×10^4–7.1×10^5)	2.2×10^6 (8.2×10^5–9.8×10^6)	2.2×10^5 (8.6×10^4–5.4×10^5)	4.7×10^7 (1.3×10^7–3.0×10^8)
334-1	7.5×10^4 (2.9×10^4–1.8×10^5)	1.2×10^7 (3.9×10^7–6.6×10^7)	4.4×10^5 (1.6×10^5–1.2×10^6)	1.7×10^8 (4.0×10^7–1.6×10^9)
334-3	2.1×10^5 (8.2×10^4–5.3×10^5)	5.1×10^7 (1.4×10^7–3.2×10^8)	8.3×10^5 (3.0×10^5–2.4×10^6)	4.5×10^9 (8.8×10^7–5.4×10^{10})
334-2	5.7×10^5 (2.1×10^5–1.5×10^6)	2.4×10^8 (5.4×10^7–2.3×10^9)	2.2×10^6 (7.4×10^5–7.8×10^6)	2.3×10^9 (3.1×10^8–5.3×10^{10})
231-1	1.3×10^6 (4.9×10^5–4.3×10^6)	9.4×10^8 (1.6×10^8–1.5×10^{10})	5.3×10^6 (1.6×10^6–2.4×10^7)	9.2×10^9 (9.3×10^8–4.5×10^{11})
231-2	3.3×10^6 (1.0×10^6–1.3×10^7)	4.2×10^9 (5.1×10^8–1.3×10^{11})	1.4×10^7 (3.8×10^6–1.0×10^8)	5.1×10^{10} (3.1×10^{10}–7.6×10^{12})
231-3	8.8×10^6 (2.4×10^6–4.8×10^7)	2.1×10^{10} (1.7×10^9–1.7×10^{12})	3.6×10^7 (7.9×10^6–4.0×10^8)	2.1×10^{11} (8.2×10^{10}–9.5×10^{13})

at concentrations 9.2×10^3 conidia/ml, 1.5×10^4 conidia/ml, 3.2×10^4 conidia/ml and 7.5×10^4 conidia/ml respectively. These results suggest that various concentrations of *B. bassiana* formulations have different efficacies to insect pests. These results are consistent with studies of the pathogenecity of *B. bassiana* against adults of different aphid species i.e., *Schizaphis graminum*, *Rhopalosiphum padi*, *Brevicoryne brassicae* and *Lipaphis erysimi* (Saif et al., 2023), maize stem borer *Chilo partellus* (Sufyan et al., 2019), and diamondback moth (DBM), *Plutella xylostella* (Agboyi et al., 2020). The differences in mortality response of various concentrations suggest that higher conidia are effective in controlling insect pests.

The study established that after 1 month, the stability of formulation based on Cfu was affected by isolate for formulation, carrier material and time. The Cfu occurred in formulation with isolates 231 and 341 formulated using DE, while highest Cfu was observed in formulation with isolate 334 formulated with either liquid paraffin or whey. After 2 months, the highest Cfu occurred in formulation with isolate 231, 341 and 334 formulated using liquid paraffin. Most of the stability after 1 month appear to respond well to oil formulations. Inorganic oils have been demonstrated to improve infection at low ambient humidity, increase the conidial tolerance to thermal stress, and enhance the conidial survival under high temperature and solar radiation (Behle and Birthisel, 2023). Dry aerial conidia, which are hydrophobic, are readily incorporated into inorganic oil formulations for subsequent spraying (Shahid et al., 2023).

5. Conclusion

In the current study, bioefficacy of seven isolates of *B. bassiana* were tested against adult and 3rd nymphal instars. *Beauveria bassiana* isolates are effective against several groups of insects pests. In the current study it took approximately 4.8 days for the best *B. bassiana* local isolate to show maximum effectiveness against the adult and 3rd nymphal instar of the desert locust. Pesticides against large swarms of desert locust need to work much faster if there is need to achieve control of migratory movements of the these insect pests. This study used three carrier materials (oils, diatomaceous earth and whey) to formulate biopesticides from *B. bassiana* isolates. The results indicate that pathogenicity was better in formulated isolates where mortality was reported at 4.2 days. The results of the formulation showed that although oil formulations was the best, whey as a carrier material which is rarely used had equally good or even better pathogenicity properties when added to some less efficacious isolates against the desert locust. The formulation with these different carrier materials also took less number of days to achieve maximal efficacy, where the best performing isolate took 4.2 days to achieve 50 % efficacy compared to 4.8 days for unformulated isolates against desert locust. The current findings reveal that while formulation with different carrier materials may improve their overall pathogenic effects, the properties of the carrier materials should be carefully considered as they may have influence on virulence properties of the isolates.

The study also revealed that different carrier materials in formulation affected the stability of the formulation, thus the current study provides encouraging data for the development of potential bio pesticides formulated with different carrier materials against desert locusts and thus useful for sustainable insect pest management.

CRediT authorship contribution statement

Pamela C Mwikali: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Lizzy A. Mwamburi:** Writing – review & editing, Supervision, Resources, Investigation,

Table 9
Statistical test of significance for the adults and 3rd stage nymphal instars for the mortality response under different concentrations of *Beauveria bassiana* isolates.

	Probit parameter estimates								Chi-square goodness of fit test			
	Adults				3rd nymphal instar				Adults		3rd nymphal instar	
Formulations	*Par. Est*	*Std. error*	*Slope (Z)*	*P-value*	*Par. Est*	*Std. error*	*Slope (Z)*	*P-value*	χ^2	*P*-value	χ^2	*P*-value
341-1	0.478	0.128	9.830	<0.0001	0.590	0.092	6.420	<0.0001	28.198	0.0004	21.555	0.0032
341-2	0.470	0.105	9.026	<0.0001	0.566	0.074	7.621	<0.0001	34.210	0.0001	30.471	0.0003
341-3	0.691	0.082	8.437	<0.0001	0.549	0.064	8.628	<0.0001	35.881	0.0001	32.811	<0.0001
334-1	0.576	0.066	8.260	<0.0001	0.493	0.059	8.310	<0.0001	27.418	0.0005	21.417	0.0032
334-2	0.487	0.059	8.250	<0.0001	0.426	0.057	7.480	<0.0001	30.158	0.0004	19.241	0.0006
231-1	0.452	0.058	7.798	<0.0001	0.396	0.057	7.004	<0.0001	24.621	0.0007	20.741	0.0045
231-2	0.413	0.057	7.295	<0.0001	0.362	0.056	6.459	<0.0001	28.911	0.0004	19.703	0.0049
334-3	0.539	0.062	8.631	<0.0001	0.470	0.059	8.301	<0.0001	26.380	0.0009	22.083	0.0002
231-3	0.378	0.056	6.744	<0.0001	0.341	0.057	6.036	<0.0001	27.601	0.0005	19.753	0.0050

Table 10
The number of *B. bassiana* Cfu in different formulations following dilutions and formulation with different carrier materials.

		Cfu at 1 month			Cfu after 2 months		
Treatments	Concentration	Liquid paraffin	DE	Whey	Liquid paraffin	DE	Whey
231	1×10^{-8}	1.32 ± 0.13	1.33 ± 0.23	1.33 ± 0.04	0.00 ± 0.00	0.00 ± 0.00	0.65 ± 0.07
	1×10^{-7}	4.02 ± 0.97	5.38 ± 0.62	4.00 ± 0.85	0.67 ± 0.05	2.00 ± 0.17	1.65 ± 0.17
	1×10^{-6}	15.62 ± 1.34	12.35 ± 1.44	9.33 ± 1.24	1.34 ± 0.11	4.33 ± 0.33	2.62 ± 0.21
	1×10^{-5}	22.02 ± 3.44	20.0 ± 2.21	17.67 ± 3.45	2.67 ± 0.145	8.67 ± 0.76	6.04 ± 0.54
	1×10^{-4}	24.32 ± 2.78	30.67 ± 3.55	27.68 ± 3.53	11.67 ± 1.09	11.06 ± 1.12	13.36 ± 1.14
	1×10^{-3}	30.02 ± 2.99	41.00 ± 3.78	35.63 ± 3.45	34.00 ± 1.34	20.36 ± 1.78	18.63 ± 1.67
341	1×10^{-8}	0.16 ± 0.02	2.33 ± 0.18	0.65 ± 0.02	0.60 ± 0.05	0.33 ± 0.04	0.34 ± 0.05
	1×10^{-7}	4.67 ± 0.42	3.33 ± 0.31	2.67 ± 0.20	1.67 ± 0.14	5.32 ± 0.44	1.36 ± 0.13
	1×10^{-6}	13.67 ± 0.12	7.67 ± 0.63	9.05 ± 0.92	4.33 ± 0.32	10.02 ± 0.99	3.00 ± 0.19
	1×10^{-5}	19.33 ± 0.15	16.67 ± 1.66	14.65 ± 1.12	11.33 ± 0.12	10.67 ± 1.22	5.67 ± 0.54
	1×10^{-4}	25.33 ± 0.23	27.67 ± 2.98	27.67 ± 1.21	24.67 ± 1.43	17.67 ± 1.42	9.33 ± 0.98
	1×10^{-3}	36.67 ± 0.24	48.00 ± 0.34	38.00 ± 2.87	34.40 ± 1.85	30.00 ± 2.76	12.33 ± 1.13
334	1×10^{-8}	0.76 ± 0.56	2.04 ± 0.22	2.38 ± 0.12	1.32 ± 0.13	1.03 ± 0.12	1.04 ± 0.11
	1×10^{-7}	3.30 ± 0.23	11.6 ± 1.35	7.68 ± 0.55	4.63 ± 0.53	3.33 ± 0.22	4.05 ± 0.42
	1×10^{-6}	12.70 ± 0.98	17.6 ± 2.33	14.34 ± 1.12	13.39 ± 0.14	6.32 ± 0.56	10.36 ± 1.03
	1×10^{-5}	21.03 ± 0.25	22.6 ± 2.12	33.03 ± 1.32	17.65 ± 0.91	11.04 ± 1.23	13.38 ± 1.15
	1×10^{-4}	34.02 ± 0.28	33.3 ± 2.84	34.34 ± 2.32	29.37 ± 1.32	16.05 ± 1.42	18.73 ± 1.65
	1×10^{-3}	47.71 ± 0.21	35.0 ± 3.01	39.06 ± 3.10	31.08 ± 1.31	27.75 ± 2.65	26.05 ± 2.02

Table 11
Tests of between-subjects effects between the treatments and persistence (cfu) measures at 1 and 2 months.

Source	Type III Sum of Squares	df	Mean Square	F	P-value
Corrected Model	14653.076[a]	105	139.553	293.102	0.003
Intercept	19466.242	1	19466.242	40884.731	0.000
Treatments	519.487	8	64.936	136.384	0.007
Carrier materials	10652.866	5	2130.573	4474.819	0.000
Time	1700.372	1	1700.372	3571.273	0.000
Treatments*Carrier materials	264.311	39	6.777	14.234	0.068
Treatments*Time	449.918	8	56.240	118.120	0.008
Treatments* Carrier materials *Time	1082.266	44	24.597	51.661	0.019
Error	0.952	2	0.476		
Total	34364.307	108			
Corrected Total	14654.028	107			

a. R Squared = 1.000 (Adjusted R Squared = 0.997)

Conceptualization. **Simon Peter Musinguzi:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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